S11. Minimal reaction implementations of a specified computational behavior

Pure Thought research portfolio

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Source: synthesis challenge 29; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

A reaction network can encode inputs as molecular amounts and produce an output through its reactions. But calling a network a computer is ambiguous unless we specify what counts as the input, the answer and an acceptable error. This problem asks for the smallest network that implements one precisely defined behavior within a bounded set of permitted reactions.

Key terms

- **Reaction network** — A set of chemical species and rules for converting reactants into products.
- **Reaction grammar** — The restrictions defining which species and reaction forms are allowed.
- **Encoding** — The mapping from abstract inputs to initial molecular states or concentrations.
- **Boolean function** — A rule mapping a finite collection of binary inputs to a binary output.
- **Deterministic semantics** — A model in which concentrations follow fixed evolution equations.
- **Stochastic semantics** — A model in which individual reaction events occur randomly.
- **Readout** — The rule and time for translating the chemical state into an output.
- **Error bound** — A limit on deviation from the required output, or on the probability of an incorrect answer.
- **Cost model** — The explicit quantity being minimized, such as reaction or species count.
- **Universality** — The ability to implement arbitrary computations under a stated encoding and resource scheme.

Why this matters

Chemical computation explores how local reactions can perform information processing, including ideas relevant to molecular sensing and control. Precise semantics make proposed constructions comparable and prevent hidden resources or flexible readout choices from creating misleading claims of simplicity.

What progress would mean

Progress would provide a verified implementation and, if claimed, prove minimality within the bounded grammar and cost model. Implementing a finite truth table would establish that task

only; computational universality requires a construction for arbitrary programs and an explicit account of resources.

Revised core question

Within a bounded reaction grammar, what is the smallest network implementing a precisely defined input/output behavior with a proved error bound?

Precise scope, target and outputs

Choose deterministic mass-action or stochastic reaction semantics, species/reaction bounds, stoichiometric arity, input encoding, output decoding and cost (species count first, then reaction count, for example). Begin with a finite Boolean input/output function and an explicit settling-time/error requirement. Enumerate networks modulo species relabeling; prove correctness over the complete allowed input set and exclude every lower-cost network within the grammar. For a universal-computation extension, supply a uniform compiler and a simulation theorem covering arbitrary program lengths, resource scaling and precision.

Detailed critique of the synthesis framing

The synthesis joins minimal autocatalysis, self-replication and universal computation, but none implies the others. "Smallest chemistry" depends on kinetics, encoding, concentration precision and which resources count as free. A fixed finite truth table does not prove Turing universality, while an analog model with unbounded precision may hide computation in its inputs. The rewrite separates a bounded minimal implementation result from a universal simulation theorem and from the RAF structure in PRD 29.

Validation and milestones

Baseline: one certified reaction implementation with explicit semantics. Strong: minimality within a finite grammar and input domain. Extension: a compiler/simulation theorem; avoid calling the baseline universal or biologically self-replicating.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$\min_{\mathcal{R}\in\mathfrak{G}} \operatorname{cost}(\mathcal{R}) \quad \text{subject to} \quad \forall x \in \mathcal{X} : \ \operatorname{error}(\mathcal{R}, x; T) \le \varepsilon.$$

Declare the finite reaction grammar $\mathfrak{G}$, semantics, encoding, decoding, time requirement, input set and cost ordering. Correctness of one network gives an upper cost bound; excluding every lower-cost network gives minimality within that grammar. A finite truth table does not prove universal computation or self-replication.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise “publication-quality” in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.